

Maria Clara Cavalcante-Siviero

PhD student

Education

2026 – Present **PhD in Astrophysics**, Max Planck Institute for Extraterrestrial Physics (MPE), Garching bei München - Germany.

Advisor: Dr. Jens Thomas

Co-advisor: Prof. Dr. Roberto Saglia

2024 – 2026 **M.Sc in Physics**, Brazilian Center for Research in Physics, Rio de Janeiro (CBPF) - Brazil.

Top ranked candidate

Highest honors

Advisor: Prof. Dr. Karín Menéndez-Delmestre

Co-advisor: Dr. Pedro Bonetti Beaklini

Master Thesis: Exploring the Underlying Universality of Dark Matter Density Profiles and the Galaxy-Halo Connection in Milky Way Analog Galaxies

Average Grade: A

2020 – 2024 **B.Sc in Astronomy with emphasis in Mathematics**, *Federal University of Rio de Janeiro (UFRJ), Valongo Observatory, Rio de Janeiro - Brazil.*

Top ranked candidate

First Class Honors

Advisor: Prof. Dr. Karín Menéndez-Delmestre

Thesis: Dark Matter Distribution in Simulated Milky Way Analogs

Average Grade: 8.7 (out of 10.0)

Research Experience

2026-present **Research project**, Advanced orbit based dynamical modeling of dark matter halos, the initial stellar mass function and Supermassive Black Holes in local galaxies and Euclid gravitational lenses

Advisor: Drs. Jens Thomas and Roberto Saglia

2025-present **Research group**, Comparing Simulations with Observations of Nearby Galaxies & Focused Stellar Populations and Implications for Supermassive Black Holes

Advisor: Prof. Dr. Karín Menéndez-Delmestre

2024-2026 **Research project**, HI observations of nearby galaxies with VLA and GMRT radio telescopes, Brazilian Center for Research in Physics (CBPF).

Advisor: Prof. Dr. Karín Menéndez-Delmestre & Dr. Pedro Paulo Bonetti Beaklini

2024-Present **Research group**, History of Star Formation in Nearby Galaxies with the CANGA Survey, Federal University of Rio de Janeiro, Brazil.

Advisors: Prof. Dr. Karín Menéndez-Delmestre & Prof. Dr. Thiago Signorini Gonçalves

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- 2021-2024 **Research project**, Comparing cosmological simulations with observations of nearby galaxies – Illustris TNG50 in focus, Federal University of Rio de Janeiro, Brazil.
Advisor: Prof. Dr. Karín Menéndez-Delmestre

Scholarships & Honors

- 2026 **IMPRS PhD fellowship at Max Planck Institute for Extraterrestrial Physics (MPE), Germany, (Accepted)**
- 2026 **Frazer Owen Fellowship and Research Assistantship at New Mexico Tech University (NMT), USA, (Declined)**
- 2025 **Research Exchange Fellowship at the NRAO Facilities in New Mexico, USA**, Studentship supported by the National Radio Astronomy Observatory (NRAO) and the National Council for Scientific and Technological Development (CNPq) for a research exchange at the VLA/NRAO Facilities, in New Mexico, USA.
- 2024-Present **MSc Scholarship at the Brazilian Center for Research in Physics**, *Master's scholarship by the National Council for Scientific and Technological Development (CNPq)*, Brazilian Center for Research in Physics, Rio de Janeiro - Brazil.
- 2021-2024 **University Scholarship**, Undergraduate research scholarship by the Brazilian Foundation for Research Support in Rio de Janeiro (FAPERJ) - extended in 2022, 2023 and 2024, Federal University of Rio de Janeiro, Brazil.
- 2024 **Honorable Mention**, 45th UFRJ Scientific Initiation Journey, Federal University of Rio de Janeiro, Brazil.

Publications

- 2025 **The Universality of Dark Matter Density Profiles for Milky Way Analog Galaxies**
Cavalcante-Siviero et al. (submitted to ApJ)
- 2025 **Dark Matter Distribution in Simulated Milky Way Analogs**
Cavalcante-Siviero & Menéndez-Delmestre. *Boletim da Sociedade Astronômica Brasileira*, 36, no. 1, 174-178

Observational Experience

- 2025B **Optical observations with the Perkin-Elmer Telescope, CANGA: Census of Austral Nearby GALaxies Survey**, Pico dos Dias Observatory, Brazil.
Responsible for the observational mission, controlling the instruments.
- 2025A **Optical observations with the Perkin-Elmer Telescope, CANGA: Census of Austral Nearby GALaxies Survey**, Pico dos Dias Observatory, Brazil.
Responsible for the observational mission, controlling the instruments.
- 2024 **Optical observations with the Perkin-Elmer Telescope, CANGA: Census of Austral Nearby GALaxies Survey**, Pico dos Dias Observatory, Brazil.
Responsible for the observational mission, controlling the instruments.

Event organization

- 2026 **1st Rio de Janeiro Extragalactic Observational School**, Brazilian Center for Research in Physics, Rio de Janeiro (CBPF) - Brazil.
Local Organizing Committee

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- 2025 **End-of-year meeting of CBPF's Cosmology and Astrophysics Coordination Group**, Brazilian Center for Research in Physics, Rio de Janeiro (CBPF) - Brazil.

Local Organizing Committee

Memberships & Academic Service

- 2026 **Member of the CBPF Postgraduate Students' Association**, Brazilian Center for Research in Physics (CBPF).
- 2024 – **Associate member of the Brazilian Astronomical Society (SAB)**
Present

Skills & Background Knowledge

- Basic VizieR, Aladin, SQL, Mathematica.
- Advanced Linux, macOS, IRAF. Python (Astropy, Scipy, Matplotlib, Numpy, etc.), $\text{\LaTeX}$, ^{3D}Barolo, CASA, CARTA, MARKOV CHAIN MONTE CARLO.

Teaching & Outreach experience

- 2025 **Science Outreach Project**, *Physicist for an afternoon*, Brazilian Center for Research in Physics (CBPF).
o Role: CBPF students mentor youth by recreating and discussing modern physics experiments.
- 2024 **Teaching assistant**, Teaching assistant for Classical Mechanics I, Physics Institute (IF/UFRJ), Federal University of Rio de Janeiro, Brazil.
- 2023 **Teaching assistant**, Teaching assistant for Mathematical Methods of Theoretical Physics I, Physics Institute (IF/UFRJ), Federal University of Rio de Janeiro, Brazil.
- 2021-2022 **Science Outreach Project**, *Astronomy for Poets* Lecture Series, Valongo Observatory (OV/UFRJ), Federal University of Rio de Janeiro, Brazil.
o Role: Effective communication of complex astronomical concepts to a broader audience.

Presentations

- 2025 **Oral presentation**, *CBPF's Astrophysics and Cosmology department end-of-year meeting*, Brazilian Center for Research in Physics, CBPF, Brazil.
- 2025 **Oral presentation**, *XLVIII Annual Meeting of the Brazilian Astronomical Society*, Brazilian Astronomical Society, Brazil.
- 2025 **Oral presentation**, *19th International Workshop on the Dark Side of the Universe (DSU2025)*, Université de Montreal.
- 2025 **Poster**, *Galaxy Memoirs*, University of Nottingham.
- 2025 **Oral presentation**, *NRAO Graduate Student Researcher Program*, National Radio Astronomy Observatory.
- 2024 **Oral presentation**, *COSMO (CBPF's Astrophysics and Cosmology department) end-of-year meeting*, Brazilian Center for Research in Physics, CBPF, Brazil.
- 2024 **Oral presentation**, *12th UFRJ Academic Integration Week*, Federal University of Rio de Janeiro, Brazil.
- 2024 **Oral presentation**, *XLVII Annual Meeting of the Brazilian Astronomical Society*, Brazilian Astronomical Society, Brazil.

- 2023 **Oral presentation**, *12th UFRJ Academic Integration Week*, Federal University of Rio de Janeiro, Brazil.
- 2023 **Poster**, *XLVI Annual Meeting of the Brazilian Astronomical Society*, Brazilian Astronomical Society, Brazil.
- 2022 **Virtual poster**, *11th UFRJ Academic Integration Week*, Federal University of Rio de Janeiro, Brazil.
- 2022 **Virtual poster**, *XLV Annual Meeting of the Brazilian Astronomical Society*, Brazilian Astronomical Society, Brazil.
- 2022 **Virtual oral presentation**, *UFRJ Scientific Initiation Day*, Federal University of Rio de Janeiro, Brazil.

Conferences, Workshops & Schools

- 2025 **Cycle of Special Courses**, *Lambda-CDM, Dark Matter, and Dark Energy - Stefano Profumo (University of California, USA)*, National Observatory, ON, Brazil.
- 2025 **Cycle of Special Courses**, *Galaxy Evolution in the Milky Way and Local Group - Carme Gallart-Gallart (Instituto Astrofisico de Canarias, Spain)*, National Observatory, ON, Brazil.
- 2025 **Cosmology and Gravitation School**, Brazilian Center for Research in Physics, CBPF, Brazil.
- 2024 **Cosmology and Gravitation School**, *XIX Brazilian School of Cosmology and Gravitation*. Brazilian Center for Research in Physics, CBPF, Brazil.
- 2024 **Workshop** *Lectures on the IllustrisTNG Cosmological Simulation*, University of São Paulo, USP, Brazil.
- 2024 **Workshop** *Galaxy Rotation and Dark Matter*, Brazilian Center for Research in Physics, CBPF, Brazil.
- 2023 **Winter School**, *Structure in the Universe, from CMB to galaxies*, Brazilian Center for Research in Physics, CBPF, Brazil.